

Milestone 4: Final Investment Memo

QM 2023 Capstone Project

Due: Friday, Week 16 (May 4, 2026) by 11:59 PM **Points:** 50 (25% of capstone grade) **Format:** Hybrid - Team memo (40 pts) + Individual addendum (10 pts)

Overview

Milestone 4 is the **culmination** of your semester-long capstone project. You will synthesize your data pipeline (M1), exploratory analysis (M2), and econometric models (M3) into a **professional investment memo** that communicates findings to a non-technical audience.

Real-world context: Investment committees don't read code or regression output. They read concise memos that translate technical analysis into actionable recommendations. Your memo must:

- State findings clearly (Executive Summary)
- Explain methods transparently (Methodology)
- Present evidence visually and numerically (Results)
- Provide investment guidance with honest caveats (Conclusions)

Success criterion: A portfolio manager with no econometrics background should be able to read your memo and make an informed investment decision.

Learning Objectives

By completing M4, you will:

1. **Translate technical results** into business language
 2. **Communicate visually** with publication-ready tables and figures
 3. **Provide actionable recommendations** grounded in empirical evidence
 4. **Acknowledge limitations** honestly (omitted variables, assumptions, external validity)
 5. **Demonstrate individual accountability** through personal contribution reflection
 6. **Practice professional writing** (clear, concise, jargon-free)
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Deliverables

1. Team Memo: [Final_Investment_Memo.pdf](#) (40 points)

Format: 5-7 pages, professional PDF (not Word doc) **Structure:** See template in [memo_template.md](#)

Required Sections:

1. **Executive Summary** (0.5 page)
 - 2-3 sentences stating the key finding
 - 1-2 sentences with investment recommendation

- No tables or figures (pure text)

2. **Methodology** (1 page)

- Data sources (REIT Master, FRED, REIT Factors) with citations
- Sample construction: n_reits, n_months, date range
- Model specifications (equations for FE and DiD/ARIMA/ML)
- Variable definitions (what is FEDFUNDS_lag2? What is MOM?)

3. **Results** (1.5-2 pages)

- **Table 1:** Fixed Effects regression (main model)
- **Table 2:** Alternative specification (DiD/ARIMA/ML)
- **Figure 1:** Key visualization (e.g., sector returns over time, rate vs. return dual-axis plot)
- **Figure 2:** Diagnostic plot (residuals vs. fitted)
- Interpretation prose: "The coefficient on FEDFUNDS (-0.025) implies that a 1 pp increase in the Federal Funds Rate reduces REIT returns by 2.5 pp, controlling for REIT and time fixed effects."

4. **Conclusions & Recommendations** (1 page)

- Investment implications: Which sectors to buy/hold/sell?
- Risk assessment: What could go wrong with this analysis?
- Caveats: Assumptions, omitted variables, external validity concerns

5. **References** (0.5 page)

- Data sources with URLs
- Any academic papers cited (Lit-Anchor from Orbis)
- Format: APA or similar

6. **Appendix: AI Audit** (0.5-1 page)

- Summary of AI use across all milestones
- Key verification and critique examples

2. Individual Addendum: **Individual_Addendum_[YourName].pdf** (10 points)

Format: 1 page PDF per student **Content:**

1. **Personal Contribution** (2-4 bullets)

- What did you contribute to M1, M2, M3, M4?
- Be specific: "Led M1 data cleaning, handled missing values and merge logic (15 hours)" not "Helped with data"

2. **One Defended Decision** (2-4 sentences)

- Choose one methodological decision you made (or advocated for) and defend it
- Example: "I recommended using a 2-month lag for FEDFUNDS based on M2 lag analysis showing strongest correlation at lag 2 ($r = -0.38$). This choice is grounded in economic reasoning: REITs negotiate leases and financing over 1-2 months, so immediate rate effects are muted."

3. **One Key Limitation** (2-4 sentences)

- What is the most important limitation or caveat of your analysis?
- Example: "Our Fixed Effects model assumes parallel trends across sectors, but M2 EDA showed slight pre-shock divergence between Retail and Industrial REITs. If this trend continued absent the rate hike, our DiD estimate may overstate the causal effect."

4. **AI Audit Notes** (if applicable)

- Any AI use specific to your work not covered in team appendix

Team Memo Template

See [memo_template.md](#) for full template. Key excerpts:

Executive Summary Example

REIT Return Sensitivity to Interest Rate Policy

Using a panel of 500+ US equity REITs over 2015-2023, we find that a 1 percentage point increase in the Federal Funds Rate reduces REIT returns by 2.5 percentage points ($p < 0.01$), controlling for REIT and time fixed effects. This effect occurs with a 2-month lag and is driven primarily by leverage and discount rate channels. Sector analysis reveals that Retail and Office REITs are highly sensitive (correlation: -0.45), while Industrial REITs are resilient (-0.15).

Investment Recommendation: Given the current elevated rate environment (5.0-5.25%), we recommend a **15% overweight** in Industrial REITs and a **neutral-to-underweight** stance in Retail and Office REITs. If the Federal Reserve cuts rates by 100 basis points over the next 12 months (as forecasted by futures markets), our model predicts a 2.5% recovery in aggregate REIT returns.

Methodology Example

Data Sources

1. **REIT Master Panel:** Monthly returns and characteristics for 532 US equity REITs (2015-2023). Source: CRSP/Ziman Real Estate Database (instructor-provided).
2. **REIT Factors:** Monthly factor premiums (SIZE, VALUE, MOM, QLTY, LOWVOL, REV). Source: Seagraves et al. (2025).
3. **Economic Indicators:** Federal Funds Rate (FEDFUNDS), 30-Year Mortgage Rate (MORTGAGE30US), CPI (CPIAUCSL), Unemployment (UNRATE). Source: FRED (Federal Reserve Economic Data).

Sample Construction

- Panel structure: 532 REITs $\times$ 108 months = 57,456 observations
- After cleaning (missing values, duplicates, outliers): 54,320 observations
- Balanced panel: No (REITs enter/exit market over time)

Model Specifications

Fixed Effects Model:

$$\text{ret_it} = \beta_0 + \beta_1 \cdot \text{fedfunds_lag2_it} + \beta_2 \cdot \text{MOM_it} + \beta_3 \cdot \text{QLTY_it} + \beta_4 \cdot \text{SIZE_it} + \alpha_i + \delta_t + \varepsilon_{it}$$

- α_i = REIT fixed effect (controls for time-invariant characteristics)
- δ_t = Time fixed effect (controls for aggregate shocks)
- Standard errors clustered at REIT level

Difference-in-Differences Model:

$$\text{ret_it} = \beta_0 + \beta_1 \cdot \text{Sensitive_i} + \beta_2 \cdot \text{Post2022_t} + \beta_3 \cdot (\text{Sensitive_i} \times \text{Post2022_t}) + \text{controls} + \varepsilon_{it}$$

- Sensitive_i = 1 if sector is Retail or Office, 0 if Industrial
- Post2022_t = 1 if month $\geq$ January 2022, 0 otherwise
- β_3 = DiD estimator (differential effect on sensitive sectors)

Results Example

Table 1: Fixed Effects Regression (Main Model)

Variable	Coefficient	Std. Error	t-stat	p-value
FEDFUNDS (lag 2)	-0.025	0.010	-2.50	0.012**
MOM	0.042	0.013	3.23	0.001***
QLTY	0.031	0.015	2.07	0.039**
SIZE	-0.008	0.011	-0.73	0.465
MCAP	0.002	0.003	0.67	0.503
REIT FE	Yes			
Time FE	Yes			
N	54,320			
R² (within)	0.38			

Notes: Clustered standard errors at REIT level. *** p<0.01, ** p<0.05, * p<0.10.

Figure 1: REIT Returns vs. Federal Funds Rate (Dual-Axis Plot)

 Dual-Axis Plot

Caption: Aggregate REIT returns (blue, left axis) exhibit inverse co-movement with the Federal Funds Rate (red, right axis). The 2022-2023 rate hike cycle coincides with declining REIT performance, consistent with our regression estimates.

Interpretation:

The coefficient on FEDFUNDS_lag2 (-0.025, $p = 0.012$) indicates that a 1 percentage point increase in the Federal Funds Rate reduces REIT returns by 2.5 percentage points, holding constant REIT and time fixed effects. This effect is:

- **Economically significant:** During the 2022-2023 rate hike (0% → 5.25%), the model predicts a cumulative 13.1% decline in REIT returns ($2.5\% \times 5.25$ pp).
- **Statistically robust:** Survives multiple robustness checks (alternative lags, exclusion of COVID crash, sector subsamples).
- **Consistent with theory:** Rising rates increase financing costs for leveraged REITs and increase the discount rate applied to future cash flows.

Factor premiums also matter:

- **Momentum (MOM):** 1 SD increase → 4.2% higher returns (trend-following profitable)
- **Quality (QLTY):** 1 SD increase → 3.1% higher returns (stable cash flows rewarded)

Size and market cap show no significant effect after controlling for factors and fixed effects.

Conclusions & Recommendations Example

Investment Implications

Based on our analysis, we provide the following portfolio recommendations:

1. Sector Allocation (Tactical):

- **Overweight Industrial REITs (15% allocation):** Low interest rate sensitivity (corr = -0.15), driven by inelastic e-commerce and logistics demand.
- **Neutral-to-Underweight Retail and Office REITs:** High sensitivity (corr = -0.45, -0.42). Expected to underperform in elevated rate environment.

2. Factor Tilts (Strategic):

- **Overweight Momentum:** 4.2% return premium per SD. Implement via trend-following signals.
- **Overweight Quality:** 3.1% return premium. Screen for REITs with high cash flow stability and low earnings volatility.

3. Rate Scenario Analysis:

- **If Fed cuts 100 bp over next 12 months:** Expect 2.5% recovery in aggregate REIT returns (with 2-month lag).
- **If Fed holds rates at 5%:** REIT returns likely to remain flat; rotate into resilient sectors.

Risk Assessment

- **Model assumes past elasticities predict future effects.** If the relationship between rates and returns has changed post-COVID, predictions may be biased.
- **Omitted variable bias:** Investor sentiment and credit conditions are unobserved. If correlated with rates, β may be confounded.
- **External validity:** Results are specific to 2015-2023 period. May not generalize to extreme rate regimes (e.g., 1980s Volcker shock).

Caveats

- **Fixed Effects assumption:** We assume time-invariant REIT characteristics are time-invariant. If REIT management quality or property portfolios change, FE may not fully control confounding.
- **Parallel trends (DiD):** Assumes sensitive and resilient sectors would have followed parallel trends absent the shock. Pre-trend divergence suggests this assumption may be violated.
- **Data limitations:** REIT factor premiums are based on tercile sorts; alternative factor construction methods could yield different results.

Individual Addendum Template

See [individual_addendum_template.md](#) for full template. Example:

Individual Addendum: Sarah Johnson

Personal Contribution:

- **Milestone 1:** Led data pipeline development (M1). Implemented FRED API integration, handled missing value imputation, and verified merge logic (20 hours).
- **Milestone 2:** Created correlation heatmap, dual-axis plot, and sector box plots. Wrote economic interpretation captions for all visualizations (15 hours).
- **Milestone 3:** Specified and estimated Fixed Effects model. Conducted heteroskedasticity and VIF diagnostics. Ran robustness checks (alternative lags, COVID exclusion) (18 hours).
- **Milestone 4:** Drafted Executive Summary and Methodology sections. Compiled team memo, formatted tables, and edited for clarity (12 hours).

Defended Decision: I advocated for using a **2-month lag** for the Federal Funds Rate based on M2 lag analysis showing the strongest correlation at lag 2 ($r = -0.38$ vs. $r = -0.20$ at lag 1). This choice is economically sensible: REITs negotiate leases and financing agreements over 1-2 months, so immediate rate effects are muted. Our robustness checks confirmed this specification: lag 2 coefficient is most statistically significant ($p = 0.012$ vs. $p = 0.08$ for lag 1).

Key Limitation: Our Fixed Effects model assumes that unobserved REIT characteristics (management quality, property mix) are **time-invariant**. However, during the COVID-19 pandemic, many REITs restructured portfolios (e.g., exited retail, entered industrial). If these changes correlate with both returns and interest rate exposure, our FE estimates may be biased. A robustness check using **two-way FE with sector-time interactions** could address this concern but requires more computational resources.

AI Audit Notes: Used ChatGPT to draft Executive Summary. Prompt: "Summarize REIT return sensitivity to interest rates for a non-technical audience." AI output was too technical (used jargon like "endogeneity" and "omitted variable bias"). I rewrote it in plain language: "Rising rates reduce REIT returns because REITs are

highly leveraged." Verified economic accuracy by cross-checking against course notes on discount rate channel.

Grading Rubric (50 points)

See [rubric.md](#) for detailed breakdown. Summary:

Component	Points	Criteria
Team Memo: Reproducibility & Rigor	10	Code runs end-to-end; models correct; diagnostics complete
Team Memo: Structure & Clarity	10	Sections organized; jargon-free; professional
Team Memo: Results & Interpretation	12	Tables/figures publication-ready; economic interpretation clear
Team Memo: Recommendations & Caveats	8	Actionable advice; honest limitations
Individual Addendum	10	Specific contribution; defended decision; key limitation; honesty

Total: 50 points

Common Pitfalls and How to Avoid Them

Pitfall 1: Executive Summary is Too Technical

Problem: "We estimated a two-way fixed effects panel regression with clustered standard errors..." **Solution:** Write for a business audience. "We analyzed 500+ REITs and found that rising interest rates reduce REIT returns."

Pitfall 2: No Investment Recommendations

Problem: Memo ends with "Results show negative correlation" (so what?) **Solution:** "Based on these findings, we recommend overweighting Industrial REITs and underweighting Retail REITs."

Pitfall 3: Ignoring Limitations

Problem: Memo claims results are definitive with no caveats **Solution:** "Our analysis assumes parallel trends, which may be violated if pre-shock trends differ. A placebo test partially addresses this concern."

Pitfall 4: Vague Individual Addendum

Problem: "I helped with coding and analysis" (no specifics) **Solution:** "I implemented the M3 Fixed Effects model, ran VIF diagnostics, and drafted the robustness checks section (18 hours)."

Pitfall 5: Tables are Unreadable

Problem: Regression output copy-pasted from Python console (raw text, no formatting) **Solution:** Use markdown tables or export to CSV → format in Excel → save as PDF

Testing Checklist (Run Before Submission)

Team Memo Checklist

- ☐ All sections present (Executive Summary, Methodology, Results, Conclusions, References, AI Audit)
- ☐ Length: 5-7 pages (not 2 pages, not 15 pages)
- ☐ Tables are formatted (no raw Python output)
- ☐ Figures are high-resolution (300 DPI)
- ☐ No jargon or unexplained acronyms
- ☐ Investment recommendations are specific (not "maybe buy REITs")
- ☐ Limitations and caveats are honest

Individual Addendum Checklist

- ☐ Personal contribution is specific (tasks + hours)
- ☐ Defended decision explains reasoning with evidence
- ☐ Key limitation is substantive (not trivial)
- ☐ Length: exactly 1 page (not 0.5, not 2)

Final Quality Check

- ☐ Team member names on both Team Memo and Individual Addendum
 - ☐ Submitted as PDF (not Word doc or Google Docs link)
 - ☐ AI Audit Appendix present (no missing = 0/50)
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Submission Instructions

GitHub Classroom Submission

1. **Accept the M4 assignment** via GitHub Classroom
2. **Add your files:**

```
git add Final_Investment_Memo.pdf
git add Individual_Addendum_[YourName].pdf # Each team member submits their
own
git commit -m "M4: Final Investment Memo submission"
git push origin main
```

3. **Verify:** Check GitHub repo; both PDFs should be visible

Deadline: Friday, Week 16 (May 4) by 11:59 PM (same day as final presentations)

Resources and Support

Office Hours

- **Dr. Seagraves:** Monday & Wednesday, 3:00-5:00 PM
- **Focus for M4:** Memo writing, investment recommendations, limitation discussions

Templates

- **Team Memo:** [memo_template.md](#) (full structure)
- **Individual Addendum:** [individual_addendum_template.md](#)

Writing Guides

- Strunk & White, *The Elements of Style* (concise writing)
- Williams & Colomb, *Style: Lessons in Clarity and Grace* (professional prose)

Presentation Integration (Week 16)

Note: M4 is due the same day as final presentations. Coordination strategy:

Wednesday (May 1): Presentation rehearsals in class **Thursday-Friday (May 2-3):** Finalize memo, incorporate presentation feedback **Friday (May 4):** Final presentations + M4 submission by 11:59 PM

Tip: Draft your Executive Summary and Conclusions sections first. These map directly to your presentation script.

Good luck! Your capstone memo is the portfolio piece that demonstrates your analytical rigor and professional communication skills.

Document prepared for QM 2023: Statistics II, Spring 2026, University of Tulsa Contact: Dr. Cayman Seagraves (cayman-seagraves@utulsa.edu)